

Unit circle



1 Calculate the fraction of the circumference of the unit circle the following angle measures represent:

a $x = 60^\circ$

b $x = 180^\circ$

c $s = 450^\circ$

2 State the quadrant in which the following angles are located:

a 299°

b 5°

c 160°

d 229°

e 40°

f 310°

g -138°

h -244°

3 Write four different angles between 0° and 360° inclusive, that lie on the quadrant boundaries.

4 For each of the following circular functions:

i State the quadrant(s) in which the function is positive.

ii State the quadrant(s) in which the function is negative.

a Sine

b Cosine

c Tangent

5 State the quadrant where the angle in each scenario is located:

a θ is an angle such that $\sin \theta > 0$ and $\cos \theta < 0$.

b θ is an angle such that $\tan \theta < 0$ and $\sin \theta > 0$.

c θ is an angle such that $\tan \theta < 0$ and $\cos \theta < 0$.

d θ is an angle such that $\tan \theta > 0$ and $\sin \theta > 0$.

6 State whether the values of the following are positive or negative:

a $\sin 31^\circ$

b $\tan 31^\circ$

c $\cos 267^\circ$

d $\sin 267^\circ$

e $\cos 180^\circ$

f $\tan 296^\circ$

g $\sin 120^\circ$

h $\cos 91^\circ$

i $\sin 296^\circ$

j $\cos 120^\circ$

k $\cos 296^\circ$

l $\sin 90^\circ$

m $\cos 51^\circ$

n $\sin 51^\circ$

o $\cos 233^\circ$

p $\tan 233^\circ$

7 For each of the following graphs, state:

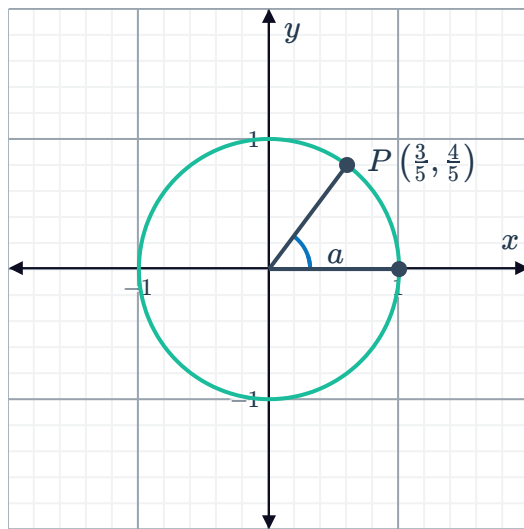


i $\sin a$

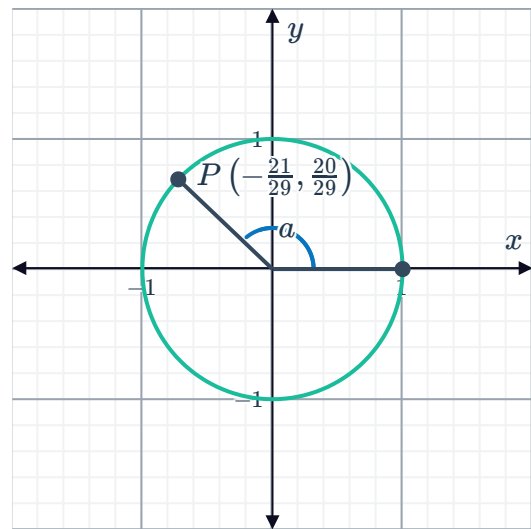
ii $\cos a$

iii $\tan a$

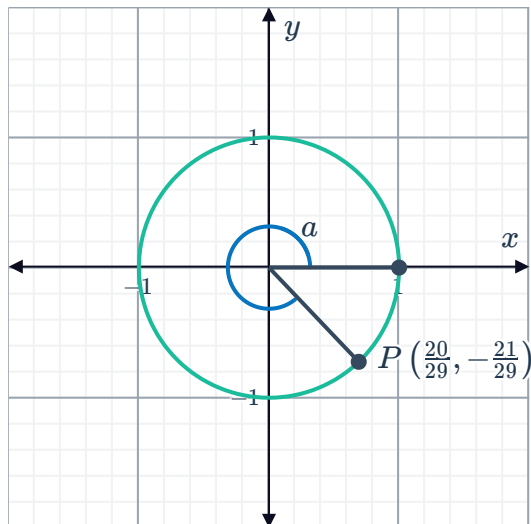
a



b



c



8 Determine whether the following statements are possible:

a $\sin \theta = \frac{7}{8}$

b $\sin \theta = -0.6$

c $\sin \theta = 1.1$

d $\sin \theta = -\frac{2}{3}$

9 Consider the following angles:



$$w = 253^\circ, \quad x = 265^\circ, \quad y = 193^\circ, \quad z = -258^\circ, \quad a = -182^\circ, \quad b = -257^\circ$$

Determine whether the following statements are true or false:

- a Angles z , a , and b are in Quadrant 3.
- b $\sin z$ and $\sin b$ are both negative.
- c $\tan(z - 90^\circ)$ and $\tan(z + 90^\circ)$ are both positive.
- d The angles w , x , and y all have negative cosine values.
- e Angles w , x , and y are in Quadrant 3.

10 Consider the following angles:

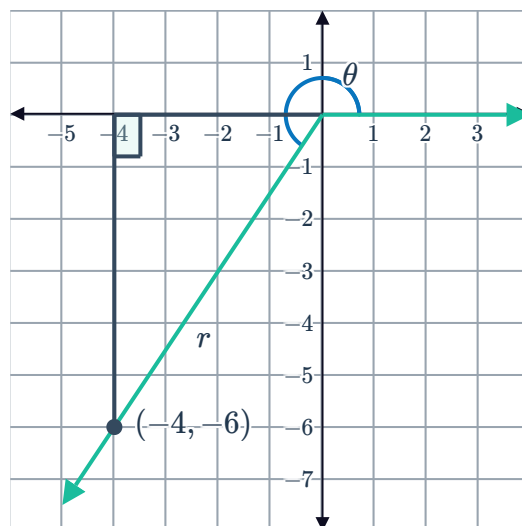
$$w = 65^\circ, \quad x = 10^\circ, \quad y = 44^\circ, \quad z = -31^\circ, \quad a = -10^\circ, \quad b = -29^\circ$$

Determine whether the following statements are true or false:

- a $\tan z$, $\cos(90^\circ + x)$ and $\sin(180^\circ + y)$ are negative.
- b $\sin x$, $\cos z$, and $\tan b$ are all positive.
- c Angles w , x , and y are in Quadrant 1.
- d $\sin x$, $\cos y$, and $\tan w$ are all positive.
- e Angles z , a , and b are in Quadrant 1.

11 The terminal side of an angle, θ passes through the point $(-4, -6)$ as shown in the diagram:

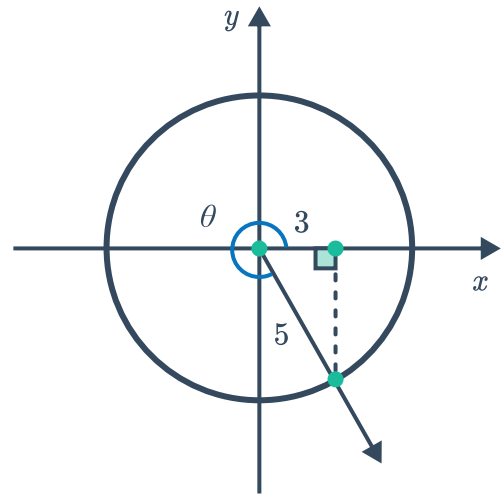
- a Find the value of r , the distance between $(0, 0)$ and $(-4, -6)$.
- b Hence find the exact values of the following:
 - i $\sin \theta$ ii $\cos \theta$ iii $\tan \theta$



- 12 Suppose that $\cos \theta = \frac{3}{5}$, where $270^\circ < \theta < 360^\circ$. Find the exact value of the following:

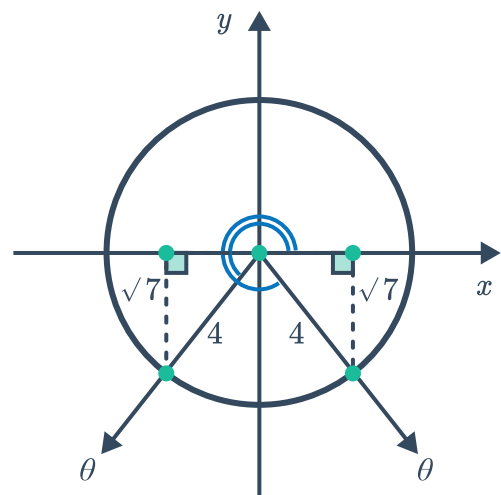


a $\sin \theta$ b $\tan \theta$



- 13 Suppose that $\sin \theta = -\frac{\sqrt{7}}{4}$. Find the exact value of the following:

a $\cos \theta$ b $\tan \theta$



- 14 Given angle θ such that $\sin \theta = 0.6$ and $\tan \theta < 0$, find the exact value of the following:

a $\cos \theta$ b $\tan \theta$

- 15 Given angle θ such that $\tan \theta = -1.3$ and $270^\circ < \theta < 360^\circ$, find the exact value of the following:

a $\cos \theta$ b $\sin \theta$

- 16 Given that $\tan \theta = -\frac{15}{8}$ and $\sin \theta > 0$, find the exact value of $\cos \theta$.

- 17 Given the following, find the exact value of $\sin \theta$.

a $\cos \theta = -\frac{60}{61}$ and $0^\circ \leq \theta \leq 180^\circ$

b $\cos \theta = -\frac{6}{7}$ and $\tan \theta < 0$

a $\cos \theta = \frac{3}{7}$ and θ is acute



b $\sin \theta = \frac{1}{\sqrt{10}}$ and $-90^\circ \leq \theta \leq 90^\circ$

Equivalent ratios

19 Consider the ratio $\sin 150^\circ$.

- a** State the quadrant in which 150° is located.
- b** State whether $\sin 150^\circ$ is positive or negative.
- c** Find the positive acute angle that 150° is related to.
- d** Hence rewrite $\sin 150^\circ$ in terms of a trigonometric ratio of a positive acute angle.

20 Consider the ratio $\cos 150^\circ$.

- a** State the quadrant in which 150° is located.
- b** State whether $\cos 150^\circ$ is positive or negative.
- c** Find the positive acute angle that 150° is related to.
- d** Hence rewrite $\cos 150^\circ$ in terms of a trigonometric ratio of a positive acute angle.

21 Consider the ratio $\tan 330^\circ$.

- a** State the quadrant in which 330° is located.
- b** State whether $\tan 330^\circ$ is positive or negative.
- c** Find the positive acute angle 330° is related to.
- d** Hence rewrite $\tan 330^\circ$ in terms of a trigonometric ratio of a positive acute angle.

22 For each of the following, rewrite the ratio as an equivalent trigonometric ratio of a positive acute angle:

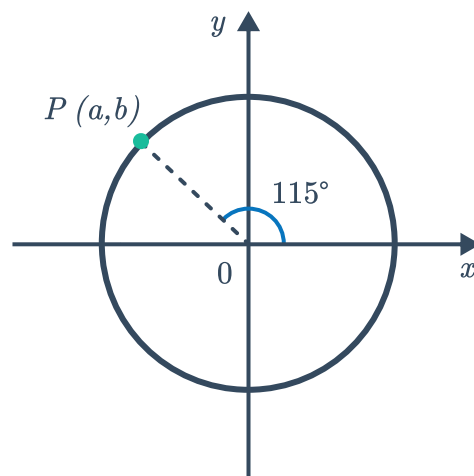
- | | | | |
|------------------------------|------------------------------|------------------------------|------------------------------|
| a $\sin 93^\circ$ | b $\cos 195^\circ$ | c $\tan 299^\circ$ | d $\sin 240^\circ$ |
| e $\sin 147^\circ$ | f $\tan (-50^\circ)$ | g $\sin 400^\circ$ | h $\cos 605^\circ$ |
| i $\tan 525^\circ$ | j $\sin 235^\circ$ | k $\cos 298^\circ$ | l $\tan 178^\circ$ |
| m $\sin (-139^\circ)$ | n $\cos (-222^\circ)$ | o $\tan (-289^\circ)$ | p $\sin (-221^\circ)$ |
| q $\cos (-103^\circ)$ | r $\tan (-40^\circ)$ | | |

- 23** Consider the following diagram of a unit circle with centre O .



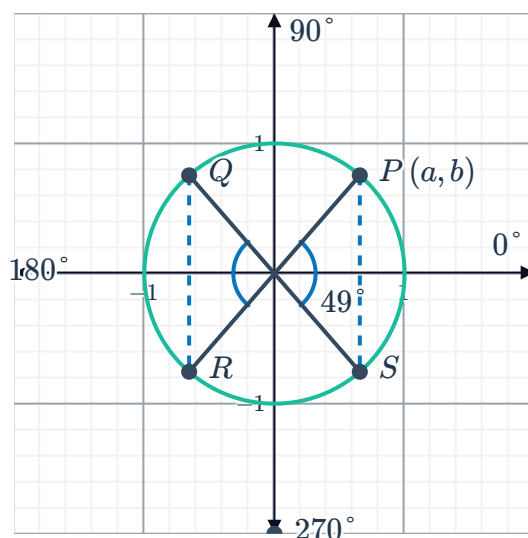
Express the following trigonometric ratios in terms of a and/or b :

- | | |
|---------------------------|---------------------------|
| a $\tan 65^\circ$ | b $\cos 475^\circ$ |
| c $\sin 65^\circ$ | d $\cos 65^\circ$ |
| e $\cos 295^\circ$ | f $\sin 245^\circ$ |



- 24** The diagram shows points P , Q , R and S , which represent rotations of 49° , 131° , 229° and 311° respectively around the unit circle.

- a** If point P has coordinates (a, b) , write the coordinates of the following points in terms of a and b :
- i** Q **ii** R **iii** S
- b** State whether the following ratios are equivalent to $\sin 49^\circ$ or $-\sin 49^\circ$:
- i** $\sin 131^\circ$ **ii** $\sin 229^\circ$
- iii** $\sin 311^\circ$
- c** Hence, express each of the following in terms of $\sin x$:
- i** $\sin(180^\circ - x)$ **ii** $\sin(180^\circ + x)$
- iii** $\sin(360^\circ - x)$



25 The diagram shows points P , Q , R and S , which represent rotations of 63° , 117° , 243° and 297° respectively around the unit circle.

- a** If point P has coordinates (a, b) , write the coordinates of the following points in terms of a and b :

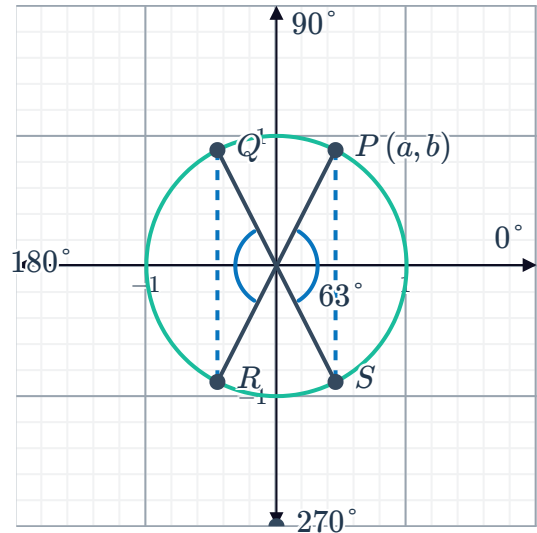
i Q **ii** R **iii** S

- b** State whether the following are equivalent to $-\cos 63^\circ$ or $\cos 63^\circ$:

i $\cos 117^\circ$ **ii** $\cos 243^\circ$
iii $\cos 297^\circ$

- c** Hence, express each of the following in terms of $\cos x$:

i $\cos(180^\circ - x)$ **ii** $\cos(180^\circ + x)$
iii $\cos(360^\circ - x)$



26 The diagram shows points P , Q , R and S , which represent rotations of 62° , 118° , 242° and 298° respectively around the unit circle.

- a** If point P has coordinates (a, b) , write the coordinates of the following points in terms of a and b :

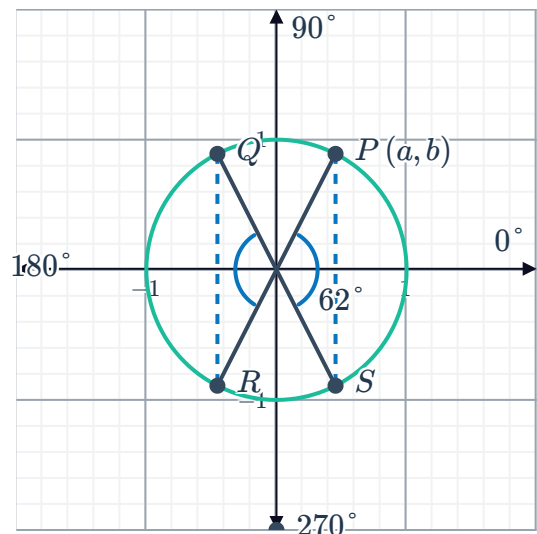
i Q **ii** R **iii** S

- b** State whether the following are equivalent to $-\tan 62^\circ$ or $\tan 62^\circ$:

i $\tan 118^\circ$ **ii** $\tan 242^\circ$
iii $\tan 298^\circ$

- c** Hence, express each of the following in terms of $\tan x$:

i $\tan(180^\circ - x)$ **ii** $\tan(180^\circ + x)$
iii $\tan(360^\circ - x)$



27 Simplify:

- | | | | |
|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| a $\sin(360^\circ + \theta)$ | b $\cos(360^\circ - \theta)$ | c $\sin(180^\circ - \theta)$ | d $\cos(180^\circ + \theta)$ |
| e $\sin(180^\circ + \theta)$ | f $\sin(90^\circ - \theta)$ | g $\sin(-\theta)$ | h $\cos(-\theta)$ |

28 Given the approximations $\cos 21^\circ = 0.93$ and $\sin 21^\circ = 0.36$, state the approximate values of the following, to two decimal places:



- | | | | |
|---------------------------|-----------------------------|-----------------------------|-----------------------------|
| a $\cos 339^\circ$ | b $\sin(-339^\circ)$ | c $\cos 159^\circ$ | d $\sin 159^\circ$ |
| e $\sin 201^\circ$ | f $\cos 201^\circ$ | g $\sin(-201^\circ)$ | h $\cos(-201^\circ)$ |

Use technology

29 Find the following, correct to two decimal places:

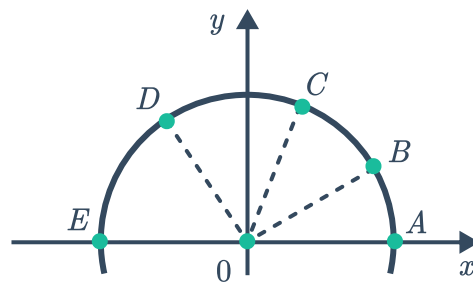
- | | | | |
|------------------------------|-----------------------------|------------------------------|-----------------------------|
| a $\sin 146^\circ$ | b $\cos 126^\circ$ | c $\tan 161^\circ$ | d $\tan 386^\circ$ |
| e $\cos 621^\circ$ | f $\cos(-218^\circ)$ | g $\sin(-42.5^\circ)$ | h $-\tan(-89^\circ)$ |
| i $(\sin 35^\circ)^2$ | j $\sin^2 35^\circ$ | k $\sin(35^\circ)^2$ | l $-\sin^2 35^\circ$ |

30 Find the following:

- | | |
|--|--|
| a $\sin^2 35^\circ + \cos^2 35^\circ$ | b $\sin^2(-64^\circ) + \cos^2(-64^\circ)$ |
| c $\sin^2 275^\circ + \cos^2 275^\circ$ | d $\sin^2 x^\circ + \cos^2 x^\circ$ |

31 Consider the following diagram of a unit circle:

- a** Given that point B , represents a rotation of 35° around the unit circle, find the coordinates of B , correct to three decimal places.
- b** Point C has coordinates $(0.294, 0.956)$. Find the angle that point C represents to the nearest degree.
- c** Given that the angle subtended by arc $\angle CD = 54^\circ$, find the coordinates of D , correct to three decimal places.



32 The diagram shows P , which represents a rotation of 66° around the unit circle. Find the following, rounding your answers to two decimal places:

- a Coordinates of point P .
- b Coordinates of point R , that represents point P reflected horizontally about the y -axis.
- c Size of the rotational angle for R around the unit circle.
- d Coordinates of point Q , that represents point P reflected vertically about the x -axis.
- e Size of the rotational angle for Q around the unit circle.

